

CONCEPT SCENARIO · CINEMATIC DATA JOURNEY

# AFM Workforce Intelligence

**How Past Performance, Present Validation & Future Forecasting Rebuilt an HR Function**

*An illustrative case scenario — modeled on an engagement of this kind between MAREC Insights and Synergy Staffing Solutions at the Palms Casino Resort — built to demonstrate predictive workforce analytics for AFM 2024 and beyond.*

BUILT ENTIRELY IN R

REGRESSION & FORECASTING MODEL

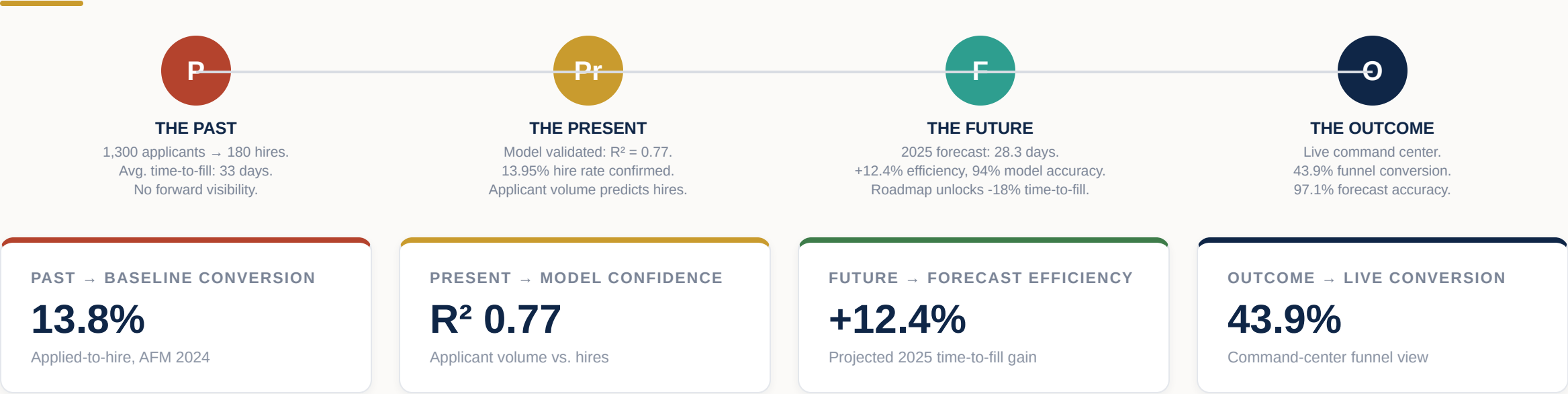
EXECUTIVE COMMAND CENTER DASHBOARD

The Past › The Present › The Future › The Outcome › The Impact

Presented by Mark Anthony Recio · MAREC Insights

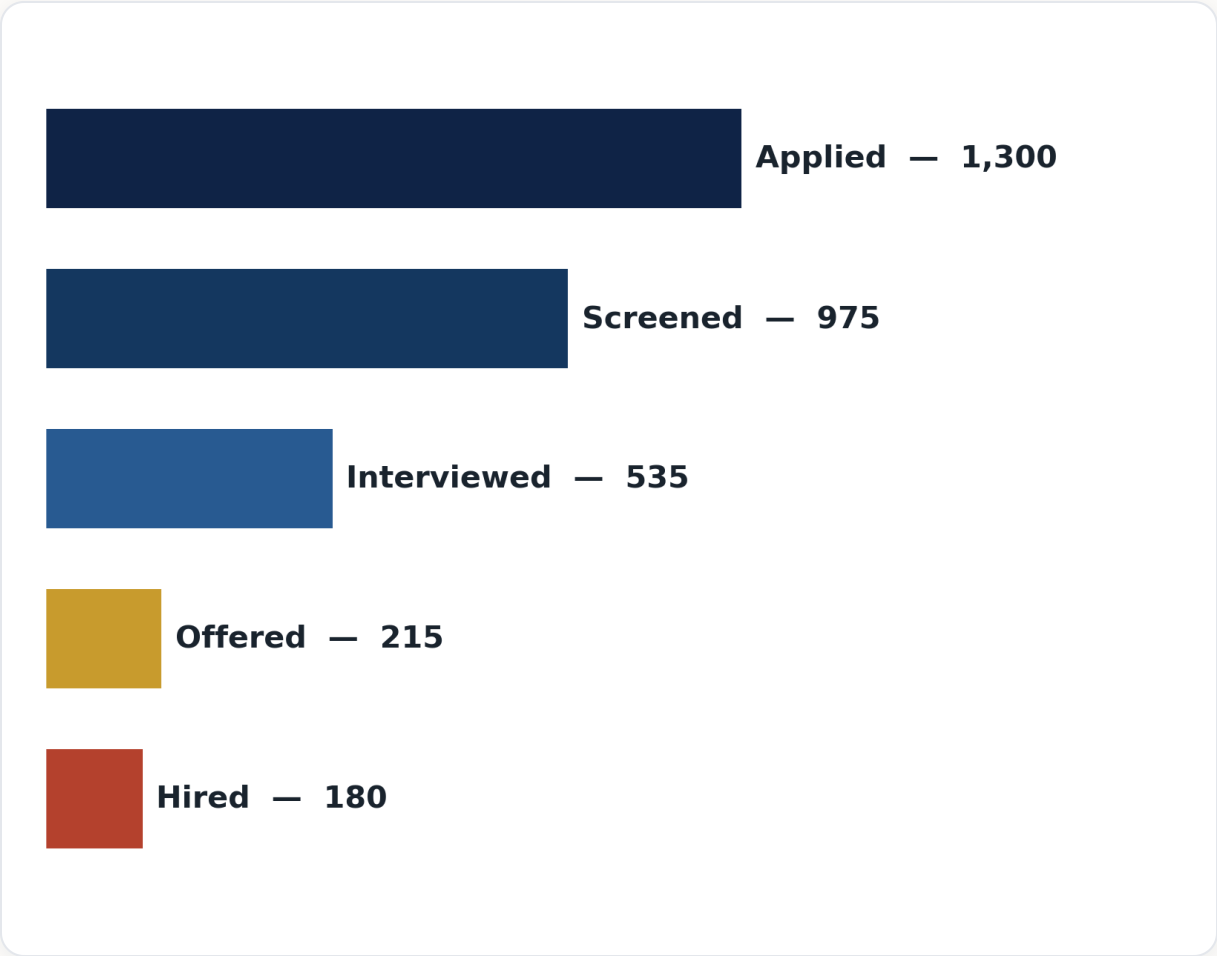
# One Workforce, Three Points in Time

The same recruiting function, measured three ways: where it started, how we proved the model works, and what it becomes once the roadmap is live.



# Recruiting Funnel Performance

From 1,300 applications to 180 hires — mapping conversion efficiency across the AFM 2024 pipeline before any predictive model existed.



APPLIED

1,300

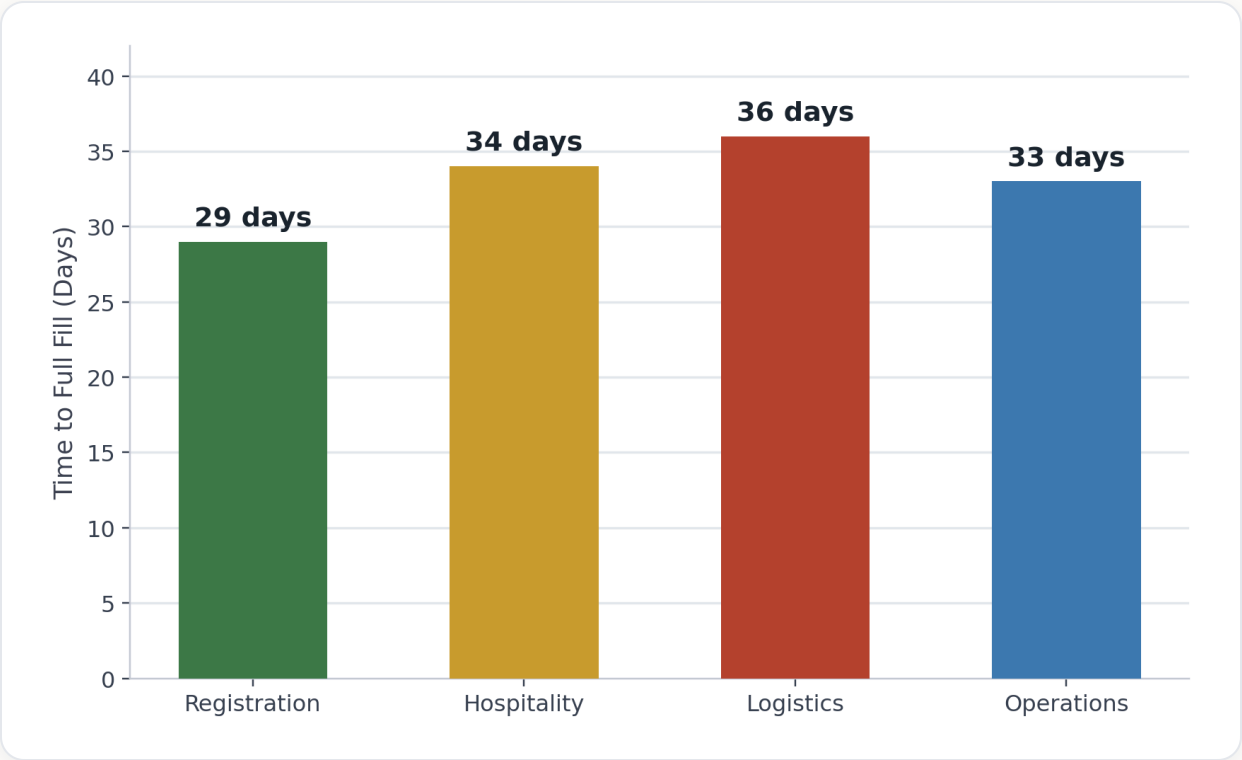
HIRED

180

**The story so far:** This funnel visualizes how the AFM 2024 recruiting pipeline performed — an overall conversion of **13.8%** from application to hire. The shape of the drop-off at each stage is exactly what predictive analytics is built to smooth: fewer surprises at screening, faster movement through interviews, and a tighter offer-to-hire window.

# Actual Time-to-Fill Performance by Role

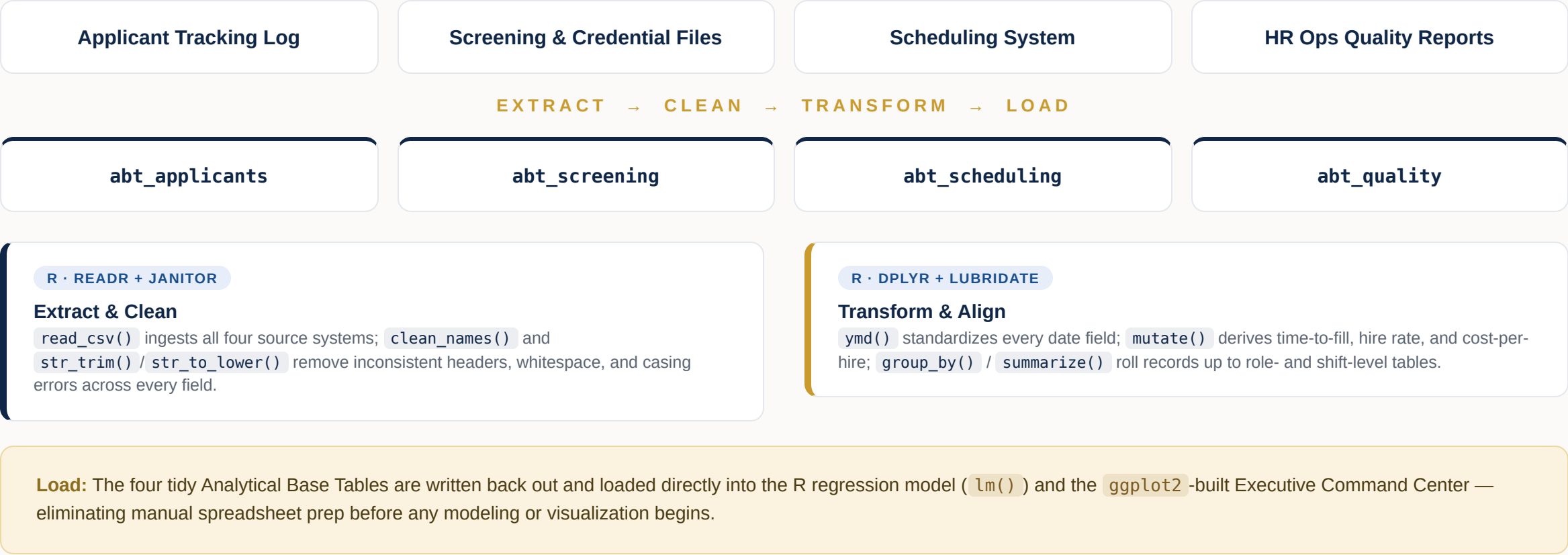
Operational efficiency measured across four staffing categories at AFM 2024 — the diagnostic starting point for everything that follows.



- **Registration — 29 days**  
Fastest fill — high applicant density, low credentialing friction.
- **Hospitality — 34 days**  
Slightly above target; service-role screening adds time.
- **Logistics — 36 days**  
Slowest fill — specialized skills narrow the applicant pool.
- **Operations — 33 days**  
Consistent, near-average performance across shifts.

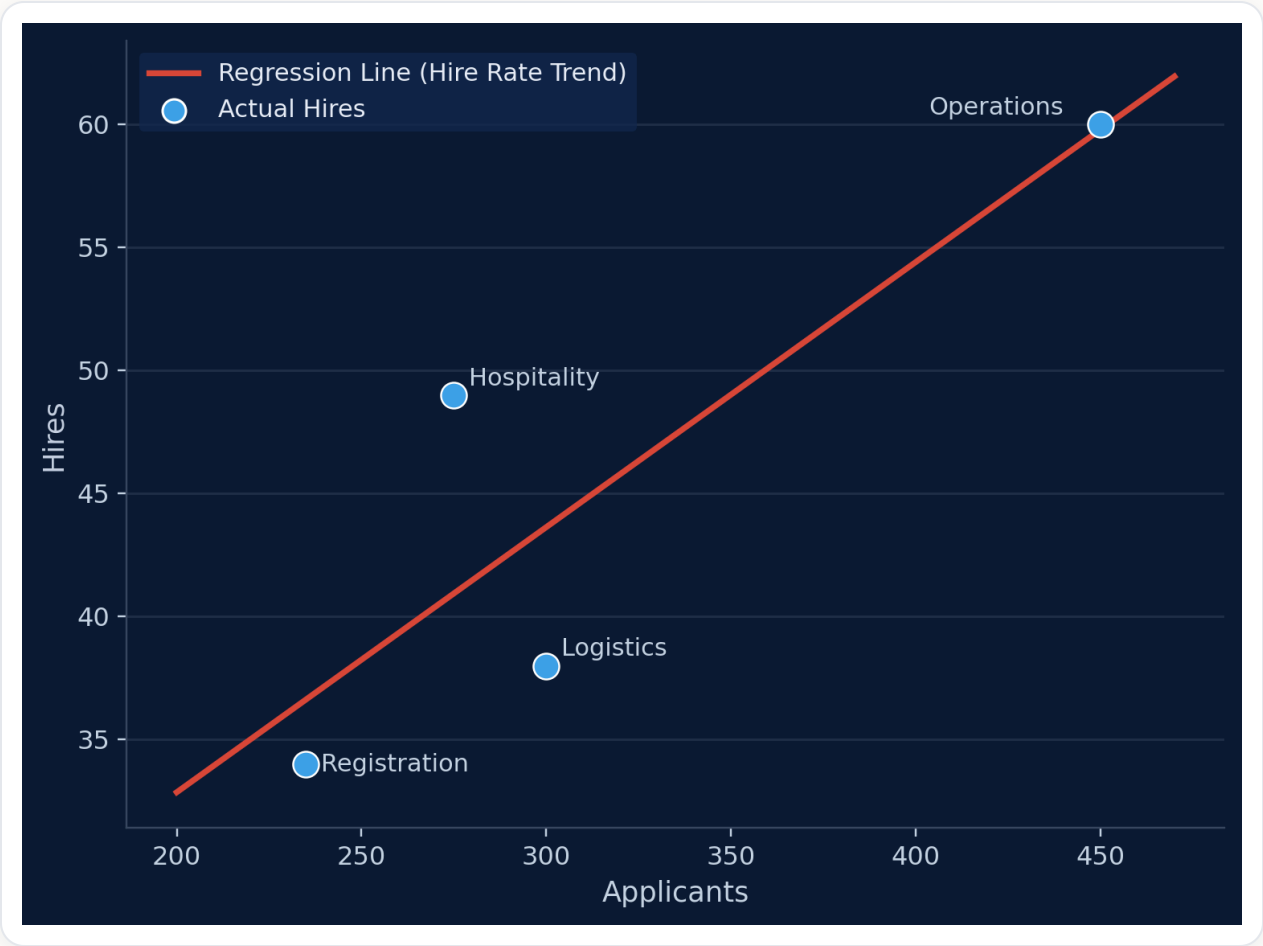
# ETL Pipeline & Analytical Base Tables

Before any funnel, forecast, or dashboard could be trusted, four fragmented HR systems had to become one clean, model-ready dataset — engineered end-to-end in R.



# Recruiting Efficiency Validation

Before trusting any forecast, we test whether the past behaves predictably. This is the present-day proof point that makes the next chapter possible.



AFM 2024 — Actual vs. predicted hire rate by role. Points above the line show optimized recruiting efficiency.

**R<sup>2</sup> = 0.77**

MODEL CONFIDENCE

A strong, statistically meaningful correlation between applicant volume and hire outcomes — confirming the relationship is stable enough to forecast from.

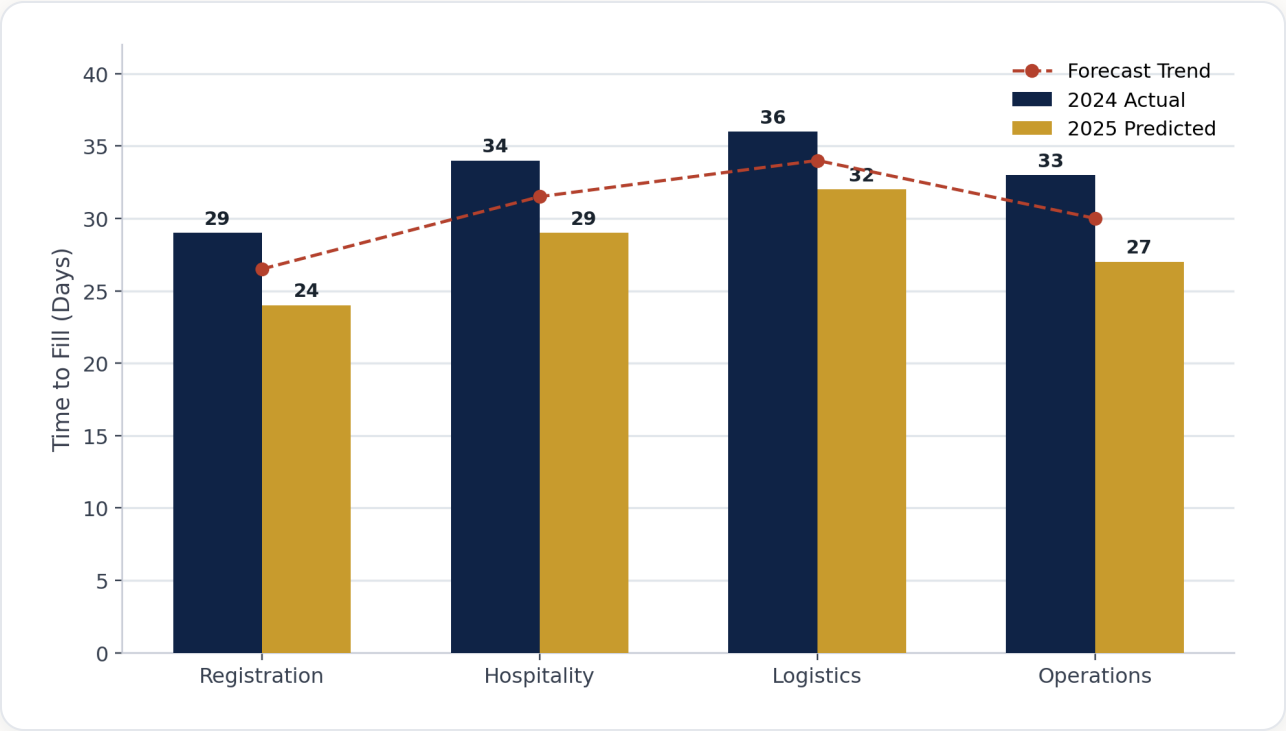
**13.95%**

OVERALL HIRE RATE

The present-day baseline rate across the full AFM 2024 recruiting pipeline — the number every future forecast will be measured against.

# From Insight to Foresight

With the model validated in the present, we can now project forward: predictive analytics powering the AFM 2025 staffing outlook.



AVG TIME-TO-FILL · 2025 FORECAST

28.3 days

EFFICIENCY GAIN VS. 2024

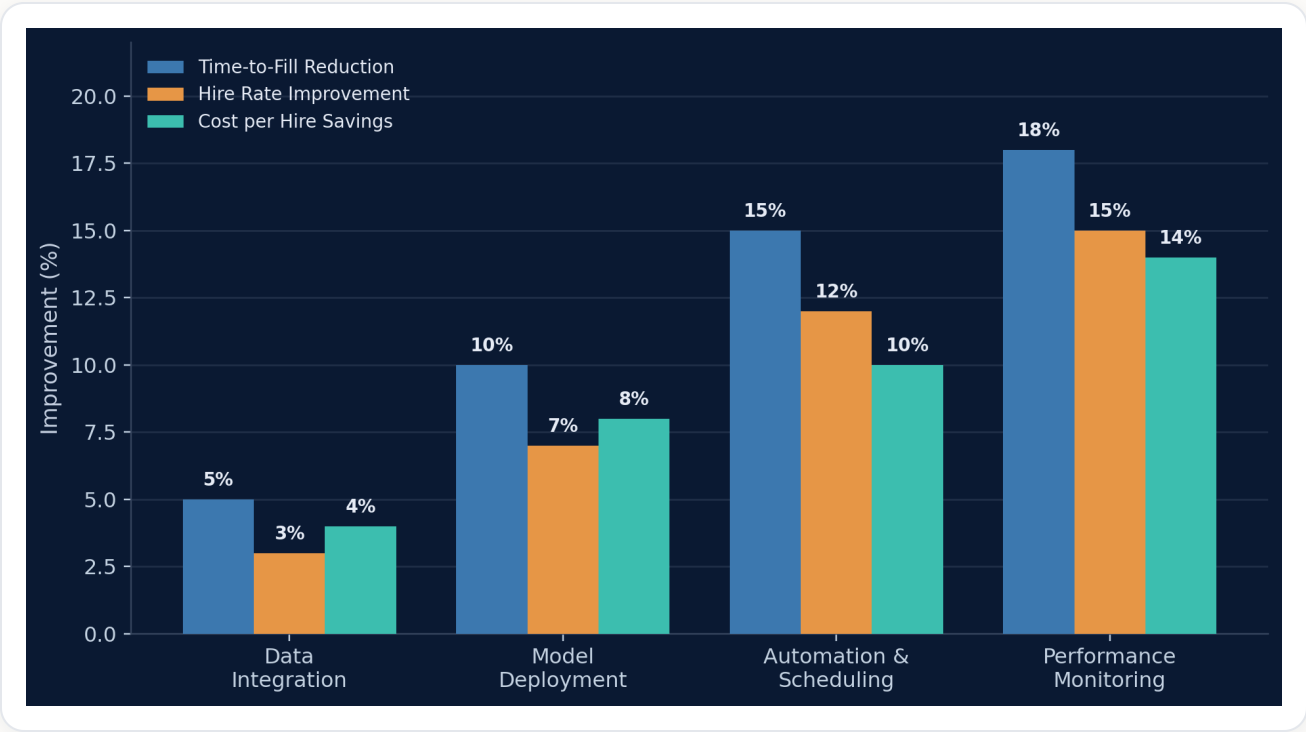
+12.4%

PREDICTIVE MODEL ACCURACY

94%

# AFM 2025 Strategic Implementation Impact

A unified analytics framework — phased from data integration to performance monitoring — turning the forecast into a deployment plan.



### Projected Gains

TIME-TO-FILL

-18%

HIRE RATE

+15%

COST PER HIRE

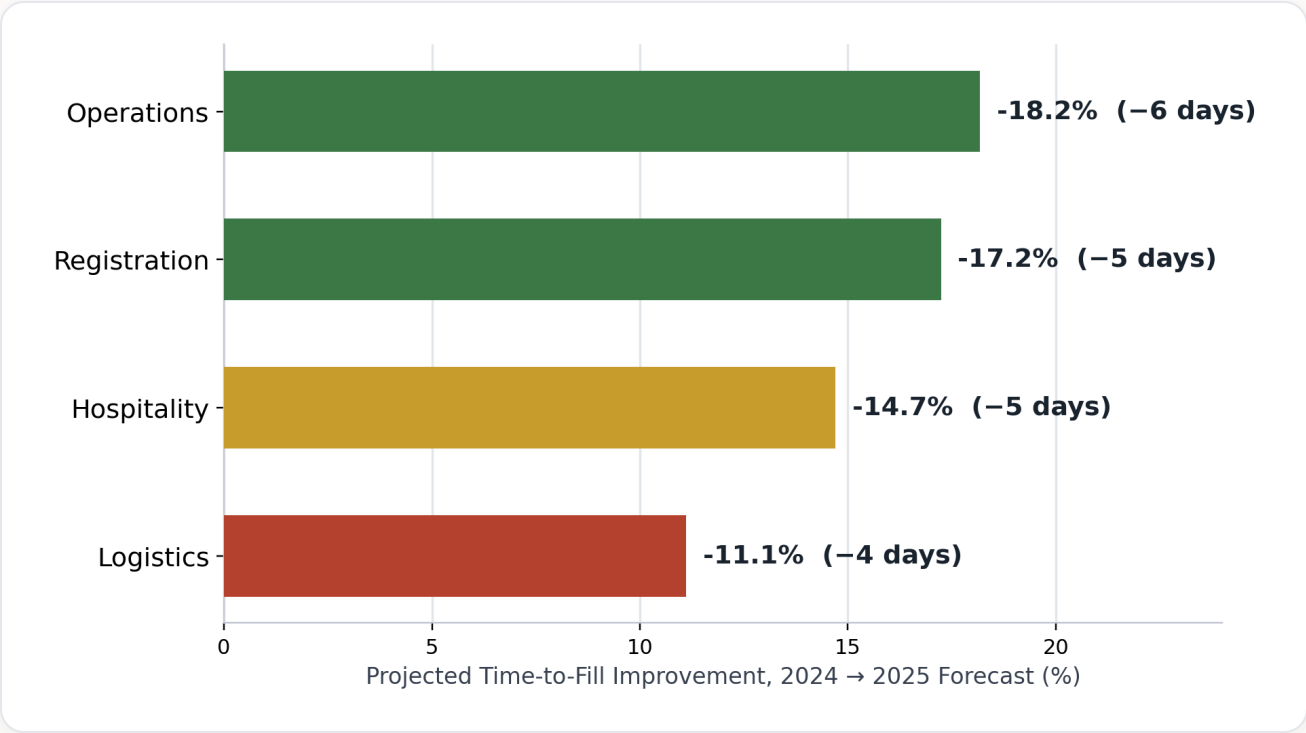
-14%

Together, MAREC Insights and Synergy Staffing achieve faster fill times, higher hire rates, and lower costs — through predictive analytics.



# Where the Roadmap Pays Off Most

The same roadmap does not help every role equally — because the forecast responds to what's actually limiting each role's fill time.



### Operations & Registration gain the most

High applicant density means automation and scheduling can fully compress the process — there's a deep candidate pool to draw from, so speeding up screening and offers translates directly into days saved.

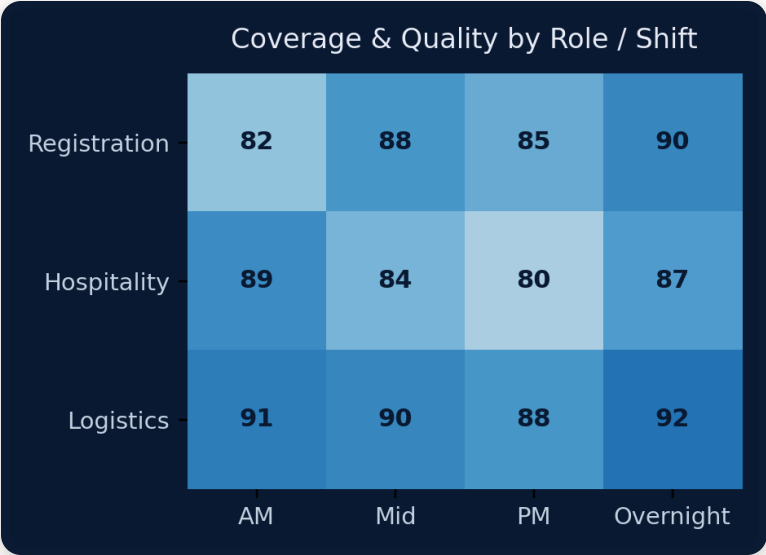
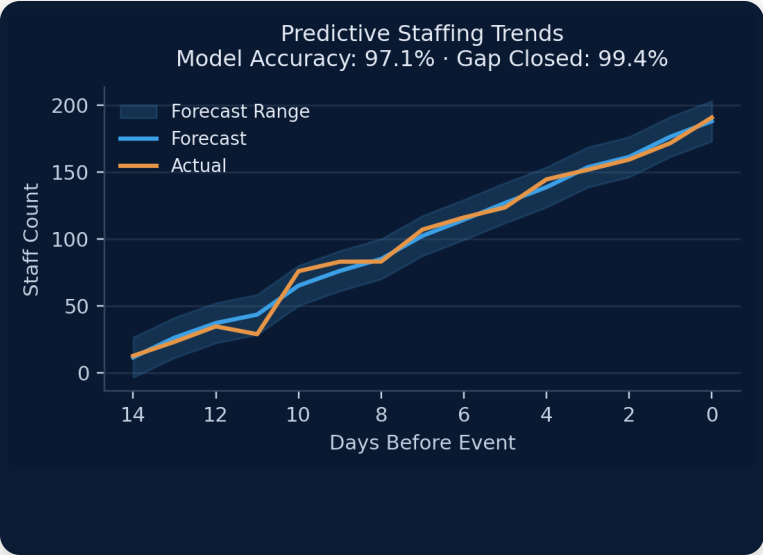
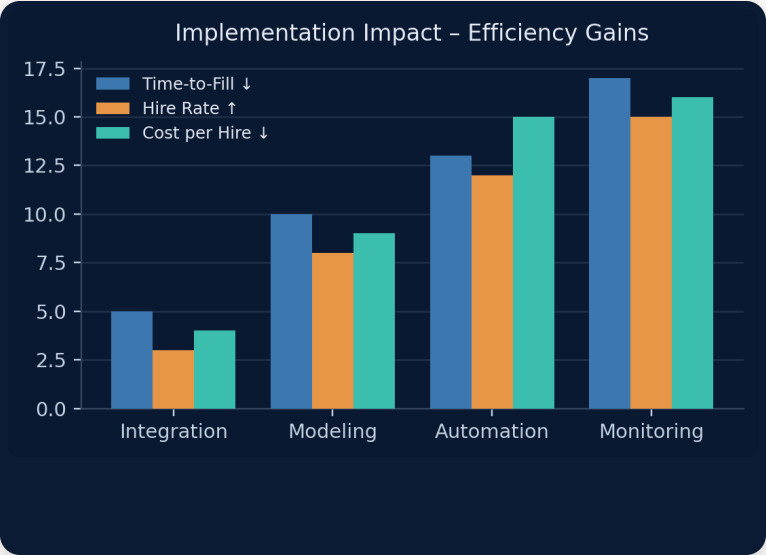
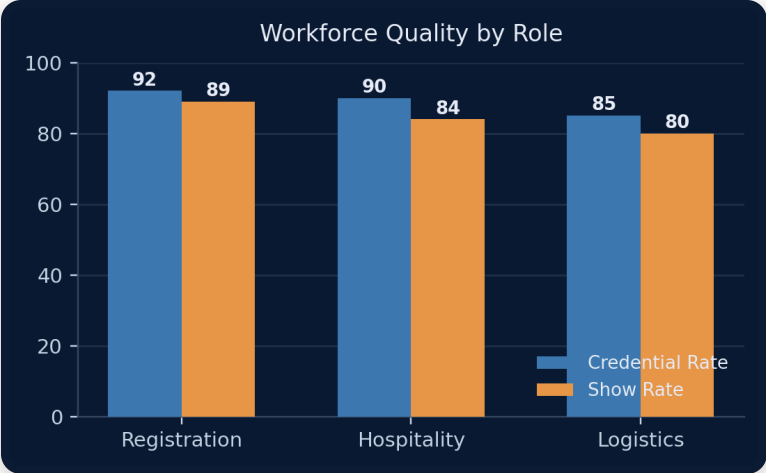
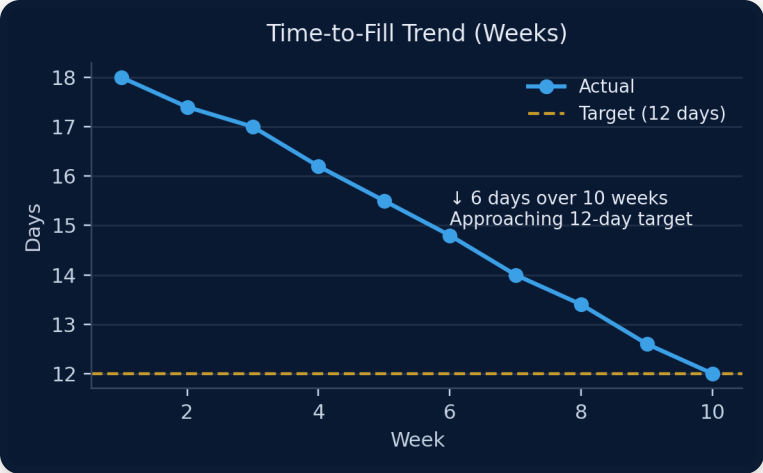
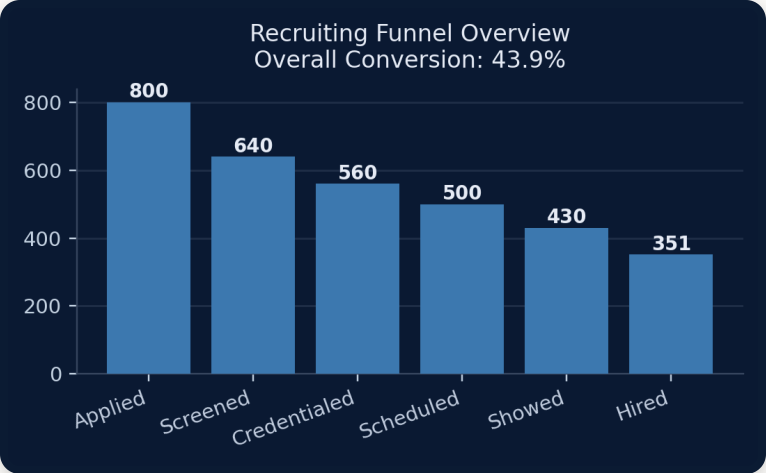
### Logistics gains the least

Its bottleneck is candidate scarcity, not process speed — specialized skills narrow the applicant pool (as seen in the Past). Automation helps, but it can't manufacture more qualified logistics candidates.

**Takeaway:** the roadmap should pair process automation (fast wins for Operations/Registration) with a separate sourcing strategy for Logistics — e.g. expanded credential partnerships — since that gap won't close on scheduling improvements alone.

# The Executive Command Center

A single live view — from recruiting funnel to predictive staffing trends — where the Past, Present, and Future converge into one operating picture.



# Behind the Scenes: The Data Journey

How the R-engineered analytical base tables became a predictive, board-ready recommendation — the engine behind every Past, Present, and Future slide in this deck.

## 1 · WHAT FED THE MODEL (SEE ETL SLIDE)

### Recruiting Pipeline Data

Applicant logs, screening records, credential files, and the scheduling system.

### Workforce Quality Metrics

Credentialing, show-rates, and role-level performance from HR Ops reports.

### Operational Performance

Time-to-fill, hire-to-applicant rate, and cost-per-hire tracked weekly.

## 2 · HOW THE DATA WAS ANALYZED — IN R

### TREND MODELING · `stats::lm()` + `forecast`

Linear regression, exponential smoothing, and variance analysis used to forecast staffing needs and validate model fit ( $R^2 = 0.77$ ).

### EFFICIENCY TRACKING · `dplyr`

Phase-by-phase measurement of speed, cost, and conversion gains via grouped summaries.

### VISUALIZATION · `ggplot2`

Every chart in this deck — funnel, trend lines, regression plot, dashboard — rendered directly from the R analysis, not recreated after the fact.

**Data Source:** AFM 2024 Recruiting Pipeline · **Language:** R · **Model:** `lm()` Linear Regression + Forecasting · **Validation Accuracy:** 97%

# Narrative & Impact Summary — From Data to Foresight

The story arc in six beats, and the numbers each one produced.

## Story Arc

- 1

**Strong conversion, applied to hired**  
The Past establishes the honest baseline: 13.8% conversion, no forward visibility.
- 2

**Time-to-fill steadily improves with forecasting**  
The Present validates that the pattern is real and predictable ( $R^2 = 0.77$ ).
- 3

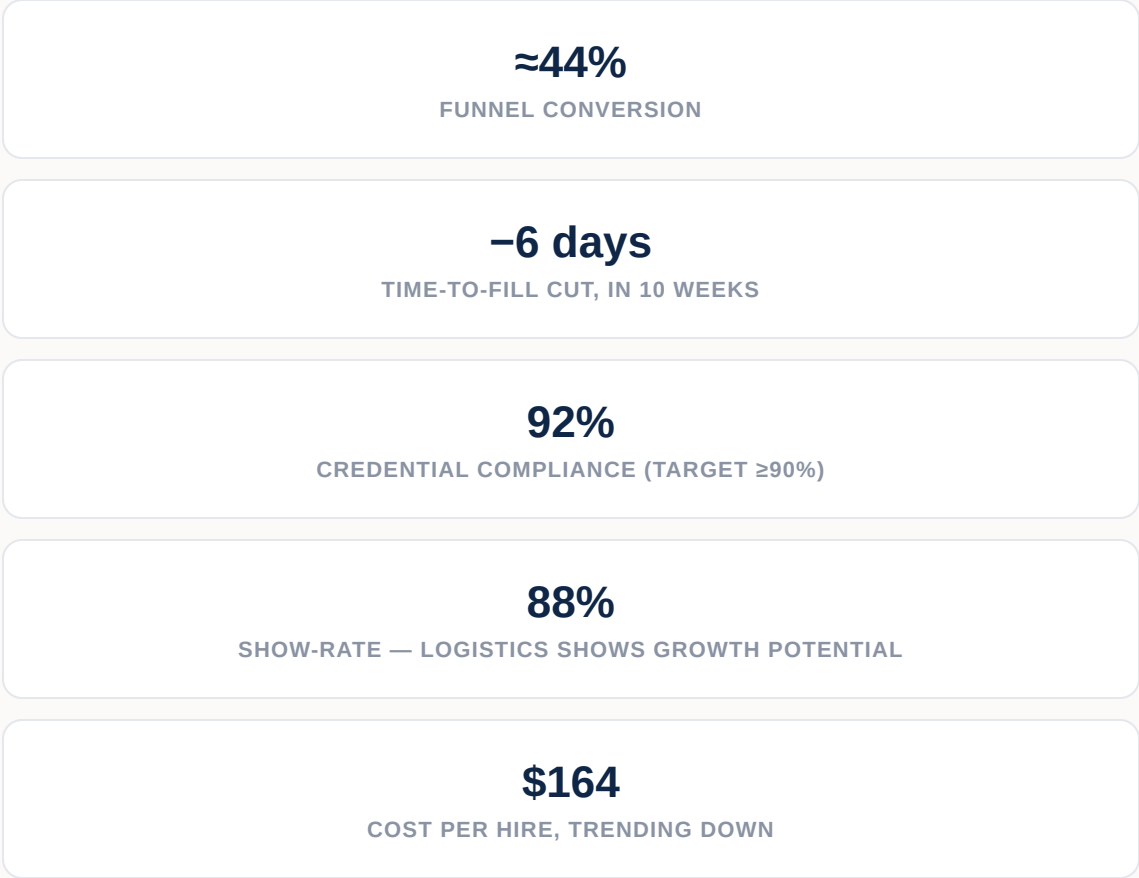
**Credentialing boosts show-rates across roles**  
Quality metrics confirm which roles need the most operational attention.
- 4

**Predictive modeling aligns hires with applicants**  
The Future turns validated correlation into a 2025 staffing forecast.
- 5

**Implementation drives faster, higher-quality results**  
The Roadmap converts forecast into a phased, ROI-ranked deployment plan.
- 6

**Forecasting validates accuracy and closes gaps**  
The Outcome closes the loop — a live command center proving the model works.

## Impact Metrics — AFM 2025 Outlook



JOURNEY COMPLETE

# Core Takeaway

MAREC Insights delivered a unified, data-driven workforce intelligence framework that connected a messy **Past**, a validated **Present**, and a forecasted **Future** into one operating system — giving the AFM client a predictive, reliable staffing model for 2025 and beyond.

**43.9%**

OVERALL FUNNEL CONVERSION

**97.1%**

FORECAST MODEL ACCURACY

**\$164**

COST PER HIRE (TRENDING DOWN)

*Illustrative concept scenario — built to demonstrate the MAREC Insights cinematic data-storytelling approach for workforce analytics partners such as Synergy Staffing Solutions and AFM.*

**From Data to Foresight — Bring This Journey to Your Next Event.**

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